

Jacob Allan — Machine Learning / Quantitative Software Engineering

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EDUCATION

University of Toronto

Toronto, Ontario

BASc in Engineering Science (3.72/4 GPA)

Expected Graduation, May 2029

- o **Concentrations:** Machine Intelligence, Mathematics | **Schulich Leader Scholar**
- o **Relevant Coursework:** Linear Algebra, Probability & Statistics, Calculus, Python/C Computing, MATLAB

SKILLS

Languages: Python, Java, C/C++, SQL, TypeScript/JavaScript, MATLAB

ML/Data: PyTorch, scikit-learn, XGBoost, NumPy, Pandas, GMMs, clustering, autoencoders, VaR, efficient frontier optimization

Tools: React, Next.js, FastAPI, Chrome Extensions, Git/GitHub, Streamlit, pytest, Ruff, Biome, Vercel, Cloud Run, LLM APIs

EXPERIENCE

Royal Military College of Canada

Kingston, Ontario

Machine Learning Researcher

May 2025 – Sept 2025

- Built and trained **5+ PyTorch CNNs** to classify **RF spectrograms** from **RTL-SDR captures** using custom Python data collection and scanning pipelines.
- Curated a **150K-sample**, 5-class RF dataset; achieved **90% average accuracy** on unseen real-world signals.
- Improved noisy-band generalization by **10% through augmentation**, learning-rate decay, frequency-offset sampling, and spectrogram normalization.
- Designed **latency-accuracy CNN variants** for embedded drone-mounted inference, supporting **RF scanning at a 2Hz** refresh rate.

University of Toronto Machine Intelligence Student Team — Flybits Industry Project

Toronto, Ontario

Machine Learning Engineer

Sept 2025 – Present

- Designing an ML-driven system to personalize digital credit offers using demographic, transactional, and product-level features.
- Built a scalable synthetic data pipeline for **100K+ customer personas** with realistic population, spending, and credit-product distributions.
- Implemented clustering and autoencoder prototypes in Python/PyTorch to create **5+ customer archetypes** and evaluate offer quality.

University of Toronto Engineering Finance Association

Toronto, Ontario

Portfolio Manager — promoted from Sales & Trading Analyst

Sept 2025 – Present

- Built a **22-factor XGBoost** return-prediction model for a **50-participant quant trading** contest using macro, technical, momentum, volatility, and rates features.
- Use **ML, VaR, and efficient frontier** optimization for asset allocation, risk management, and trade proposals.
- Develop thesis-driven trade proposals combining model outputs, risk constraints, transaction-cost assumptions, and macroeconomic research.

PROJECTS

Regime Specialist Stock Predictor | Python, XGBoost, GMM, Streamlit |

Jun 2023 – March 2025

- Built a regime-aware trading evaluation system using **GMM** clustering and **16 XGBoost** return-prediction models.
- Engineered **10+ time-series indicators**; achieved **70% directional accuracy** and **60% return accuracy** on one year of unseen data.
- Created a Streamlit app for real-time regime classification, model comparison, and trade evaluation.

Incident Triage Copilot | Next.js, FastAPI, Groq, LLM APIs, Cloud Run, Vercel |

Mar 2026 – Apr 2026

- Built an **AI incident triage app** that converts alerts, logs, deployment notes, service context, and metrics into severity, root-cause, next-action, and confidence outputs.
- Implemented typed API contracts, pytest coverage, **Groq/Gemini provider support**, **Vercel frontend** deployment, and Cloud Run backend deployment.

FormatClip | Chrome MV3, React, TypeScript, FastAPI, LLM APIs |

Feb 2026 – Present

- Built a Chrome extension and **FastAPI backend** for saving snippets, formatting selected text, and reusing cleaned outputs with provider-swappable **LLM support**.
- Tested copy, replace, delete, clear, persistence, and backend formatting workflows across the Chrome side panel and FastAPI service.